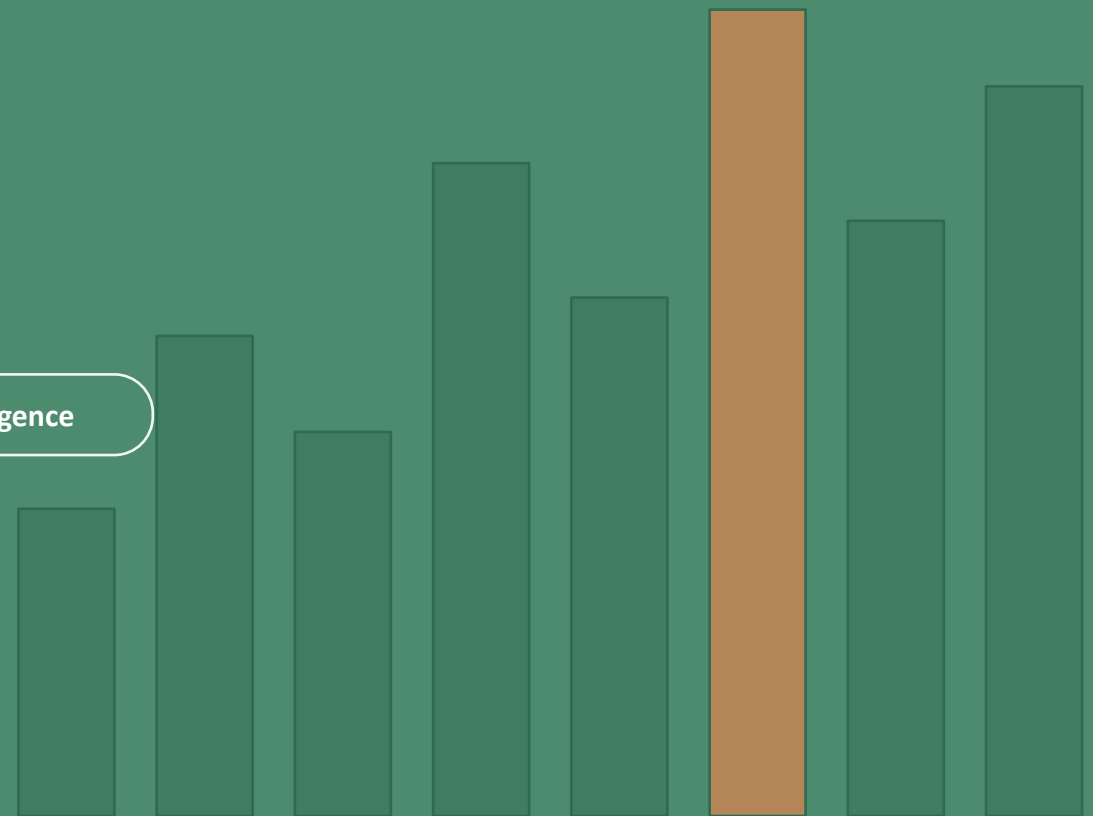
PORTFOLIO CASE STUDY

Hotel Performance Analytics

Turning Operational Data into Business Insights

SQL | Data Warehousing | Power BI | Business Intelligence

Reporting period: April 22, 2022 – July 31, 2026



Revenue and Realized Room Rates Fell About 15–16% Between 2023 and 2025

REVENUE

−16.1%

SAR 1,373,160 (2023) → SAR 1,152,561 (2025)

ADR

−15.2%

SAR 330.98 (2023) → SAR 280.65 (2025)

CANCELLATION RATE

34.35%

Peak in 2024; 29.34% in 2023 and 33.04% in 2025

Revenue pressure coincided with modestly fewer completed bookings and, more strongly, lower realized room rates. Cancellations peaked in 2024 and remain above 2023. These are descriptive patterns, not proof of cause.

Managers Need One Integrated View of Revenue, Demand, Pricing and Cancellations

THE BUSINESS QUESTION

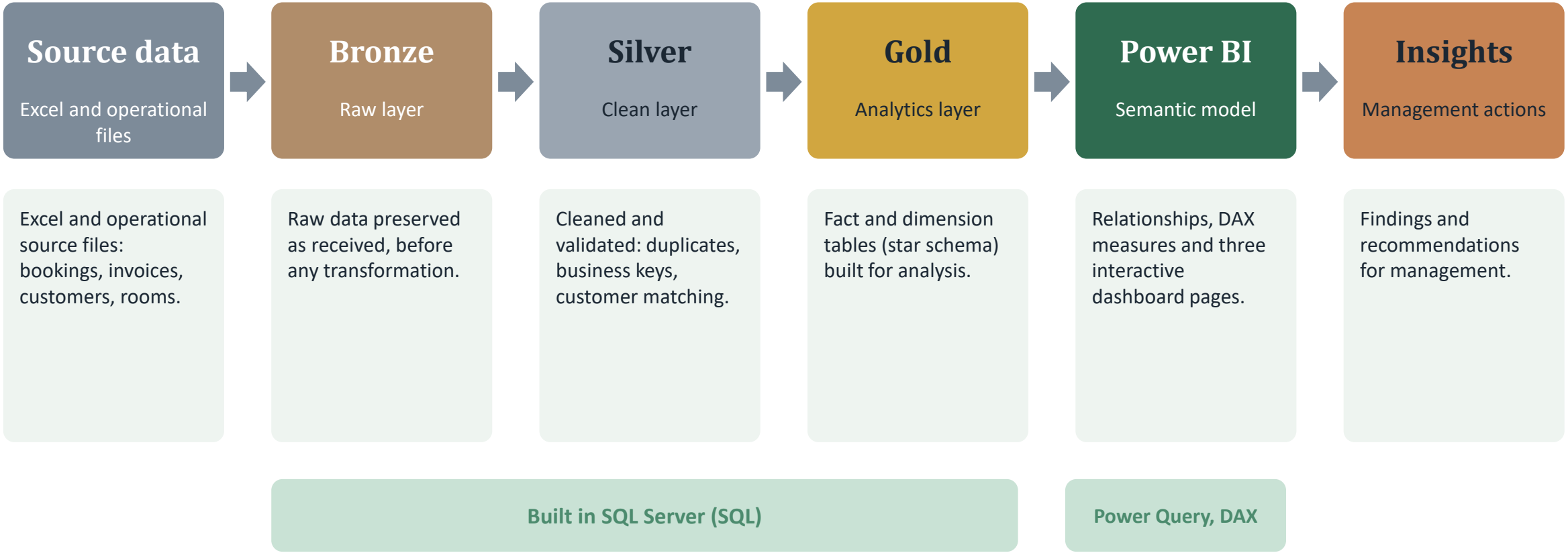
How can hotel management identify the main patterns behind changing revenue and make better-informed commercial decisions?

1. **Revenue**
   The headline outcome, but it cannot explain itself.
2. **Bookings**
   How much demand converts into completed stays.
3. **Occupancy**
   How much of the available capacity is used.
4. **Room pricing (ADR, RevPAR)**
   The rate achieved and revenue per available room.
5. **Cancellations**
   Demand that does not convert, and when it happens.

Seven Business Questions Drive the Metrics and the Story

#	Business question	Metrics used	Slide
1	How did revenue and completed bookings change over time?	Revenue, completed bookings, revenue per completed booking	9
2	Which room categories contribute the most revenue?	Revenue by room category	10
3	Which categories achieve the highest occupancy, ADR and RevPAR?	Occupancy rate, ADR, RevPAR by category	10
4	Is the decline linked to bookings, realized rates, or both?	Completed bookings, ADR, occupied room nights	9, 11
5	How has the cancellation rate changed?	Cancellation rate by year	12
6	Does revenue show recurring monthly patterns?	Monthly revenue	13
7	What could management investigate to improve performance?	KPIs tied to each recommendation	14

Raw Files Become Decision-Ready Dashboards Through a Layered Pipeline



Data Was Validated Before Modeling: Duplicates, Keys and Relationships

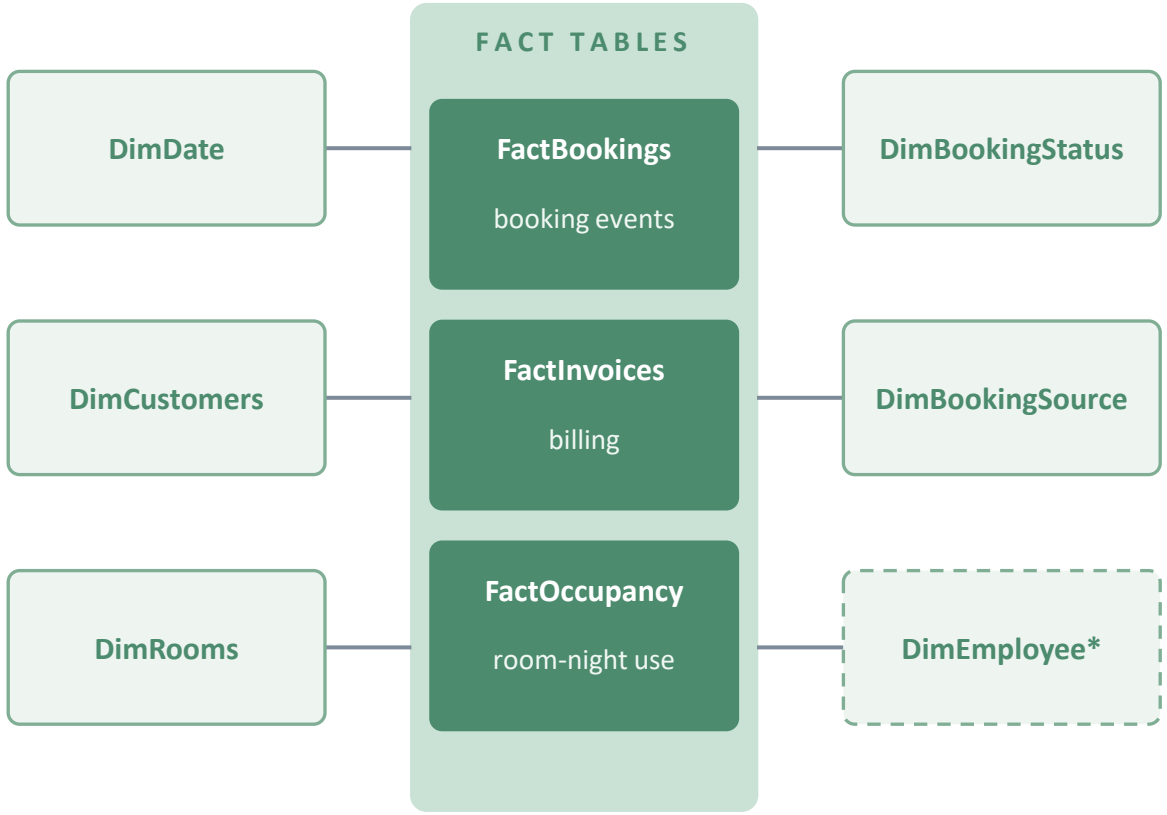
26 duplicate booking records identified

8 duplicate invoice records identified

Historical validation findings during investigation, not a statement about the final dataset.

1. **Duplicate bookings and invoices**
   Investigate first: remove confirmed duplicates, preserve legitimate records.
2. **Booking numbers reused across years**
   Validate a composite business key (for example booking number plus year).
3. **Customer matching across tables**
   Check that customers match across the source tables.
4. **Fact-to-dimension relationships**
   Confirm keys and relationships before building the model.
5. **Table grain and measure consistency**
   Define each fact table's grain so measures aggregate consistently.

A Star Schema Keeps Every KPI Calculated from One Consistent Model



*Illustrative concept. Each analysis uses the relevant subset of tables. *DimEmployee only if in the final model.*

KPI	Plain-language definition
Revenue	Realized revenue in SAR for the selected period
Total Bookings	All booking records, any status
Completed Bookings	Bookings with completed status
Cancellation Rate	Cancelled bookings ÷ total bookings
Occupancy Rate	Occupied ÷ available room nights
ADR	Room revenue ÷ occupied room nights
RevPAR	Room revenue ÷ available room nights

Conceptual definitions; exact DAX not shown.

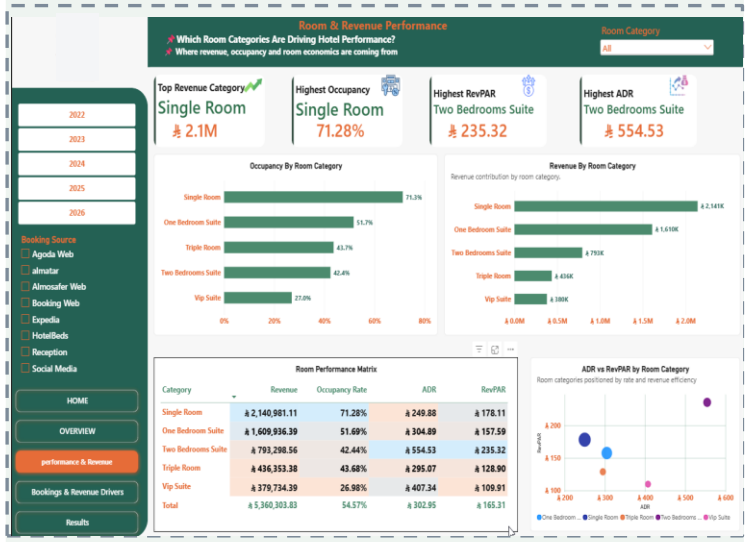
Three Dashboard Pages Move from Overview to Room Detail to Revenue Drivers

1 Hotel Performance Overview



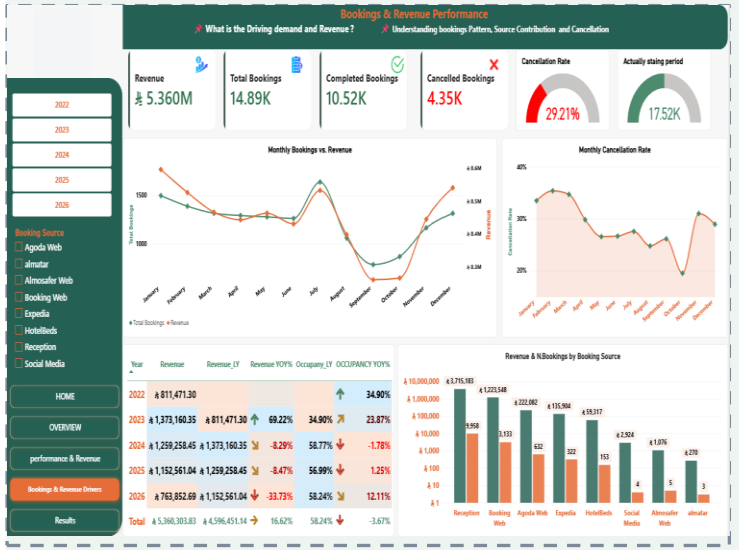
Headline revenue, bookings and cancellation KPIs over time.

2 Room & Revenue Performance



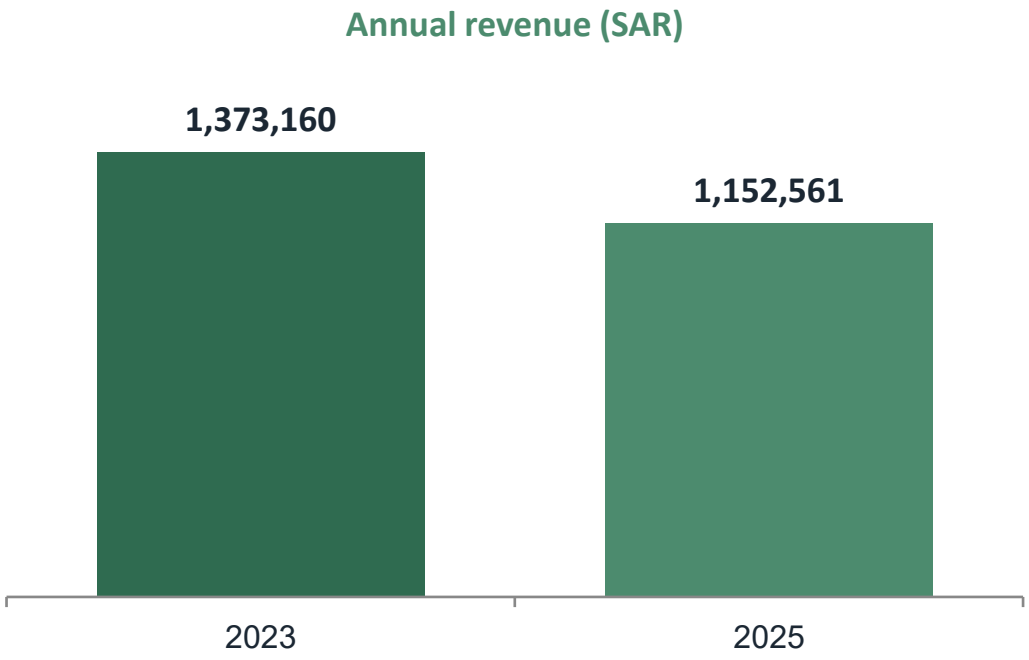
Revenue, occupancy, ADR and RevPAR by room category.

3 Bookings & Revenue Drivers



Bookings, ADR and occupied room nights behind revenue changes.

Hotel Revenue Declined by 16.1% Between 2023 and 2025



Revenue

SAR 1,373,160 → SAR 1,152,561

−16.1%

Completed bookings

2,537 → 2,361

−6.9%

Revenue per completed booking

SAR 541.25 → SAR 488.17

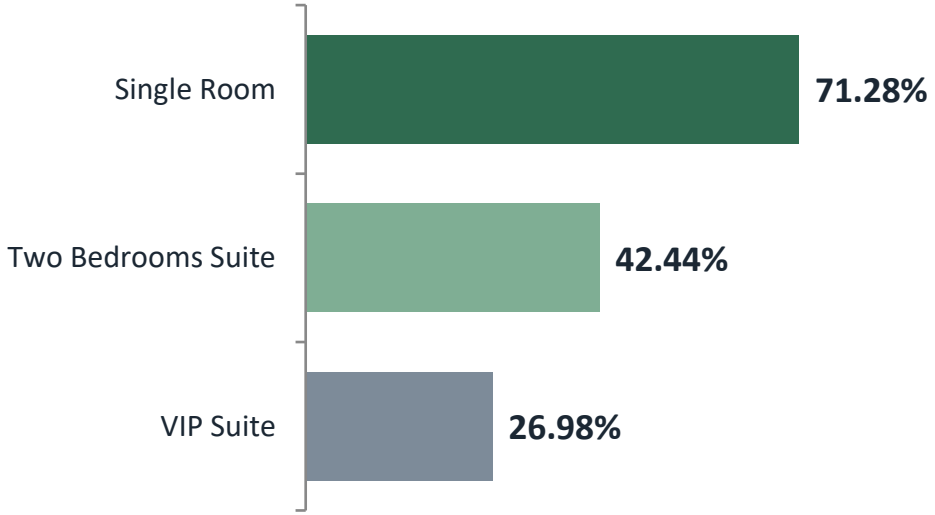
−9.8%

Fewer bookings (−6.9%) combined with lower revenue per completed booking (−9.8%) account for the −16.1% revenue change.

Revenue per completed booking is a descriptive metric, not ADR. Numerator and denominator should cover compatible populations.

Single Room Brings Volume While Suites Bring Rate

Occupancy rate by room category (%)



The largest revenue category is not necessarily the one with the highest rate or revenue per available room.

Category	Revenue	Occupancy	ADR	RevPAR
Single Room	≈ SAR 2.14M	71.28%	n/s	n/s
Two Bedrooms Suite	n/s	42.44%	SAR 554.53	SAR 235.32
VIP Suite	n/s	26.98%	n/s	SAR 109.91

Single Room

≈ SAR 2.14M revenue at 71.28% occupancy: the volume engine.

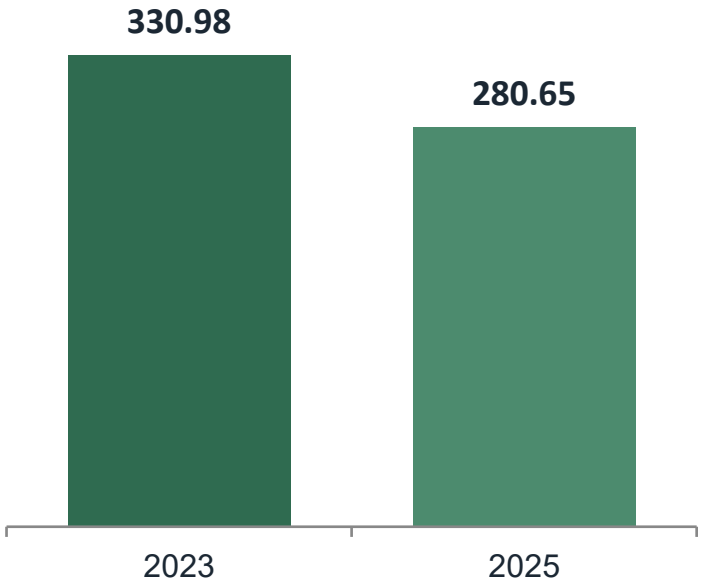
Two Bedrooms Suite

Higher ADR and RevPAR, but occupancy of only 42.44%.

Overall period Apr 22, 2022 to Jul 31, 2026. n/s = not supplied for this deck; add from the Room & Revenue dashboard page.

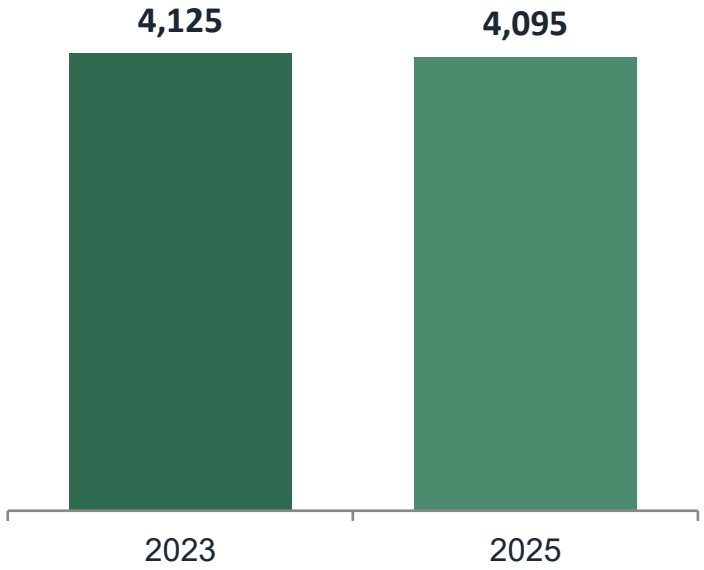
Revenue Pressure Coincided with Lower Realized Room Rates

ADR (SAR per occupied room night)



ADR −15.2%

Occupied room nights



Nights −0.7%

ADR change by category, 2023 → 2025

−13.8%

Single Room

−19.4%

One Bedroom Suite

To investigate, not conclusions: discounting, booking-source mix, room-category mix and length of stay.

Cancellation Rate Peaked at 34.35% in 2024 and Stayed Above 2023

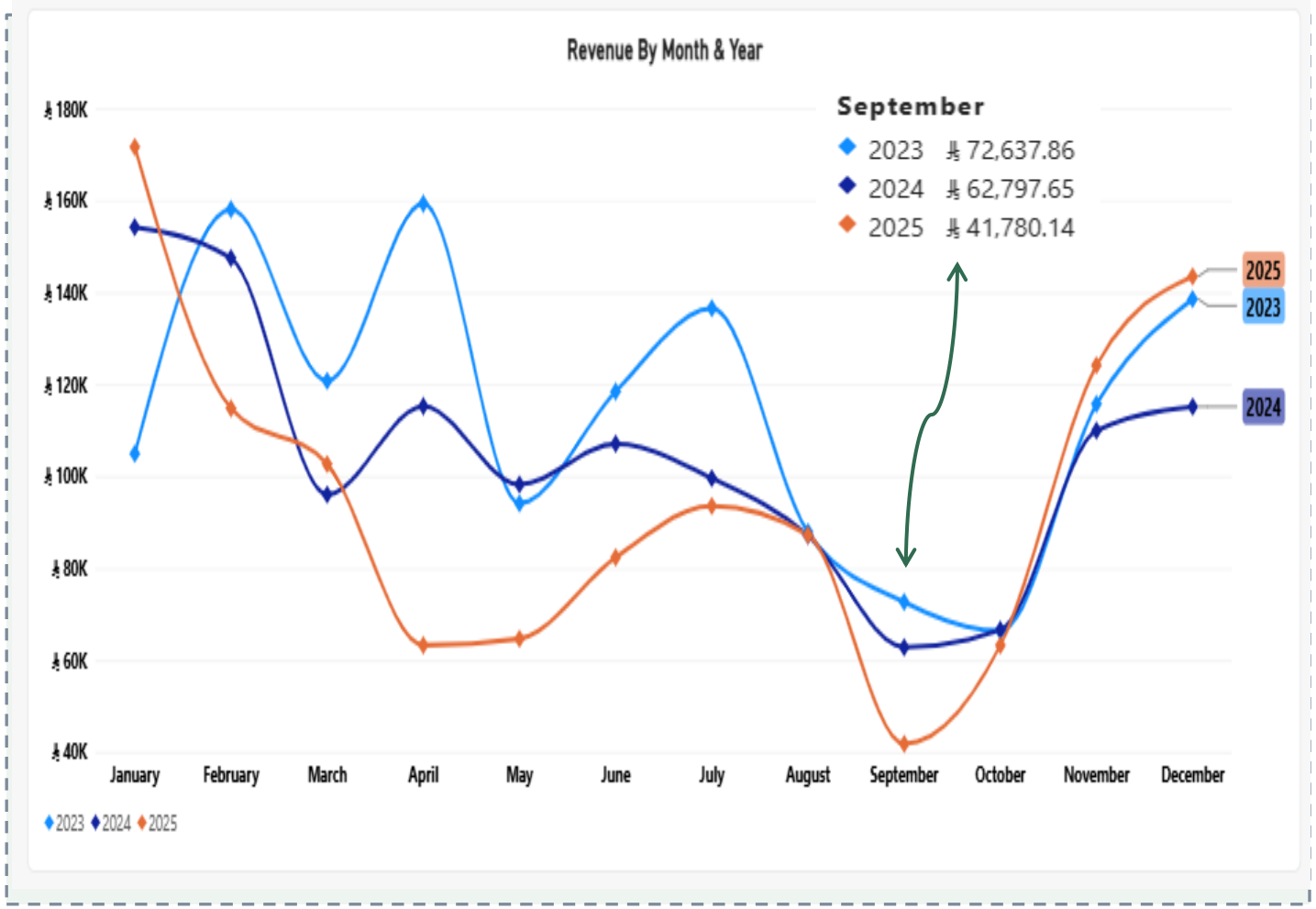


2023 → 2024	+5.01 pp
2024 → 2025	–1.31 pp
2025 vs 2023	+3.70 pp

A cancellation is not lost revenue. Timing, policy terms and rebooking still need review.

Changes shown in percentage points (pp), not percent.

September Shows a Recurring Revenue Dip in 2023, 2024 and 2025



OBSERVED

Monthly revenue dips in September in each of 2023, 2024 and 2025.

INTERPRETATION

Consistent with lower demand, which operating experience also supports. Revenue alone does not prove the cause.

ACTION

Plan ahead: pricing review, targeted promotions, occupancy planning and cost alignment.

Five Areas for Management to Investigate, Each Tied to a KPI

#	Recommendation	Supporting evidence	KPI to monitor
1	**Investigate the decline in realized room rates**	ADR fell 15.2% (SAR 330.98 → 280.65) while occupied room nights fell 0.7%	ADR by category and month; booking-source mix
2	**Review Single Room and One Bedroom Suite performance**	ADR down about 13.8% (Single) and 19.4% (One Bedroom Suite), 2023 → 2025	ADR, RevPAR and occupancy for both categories
3	**Evaluate ways to lift Two Bedrooms Suite occupancy**	ADR SAR 554.53 and RevPAR SAR 235.32, with occupancy of 42.44%	Occupancy and RevPAR for the suite
4	**Review cancellation patterns and policies**	Rate rose from 29.34% (2023) to 34.35% (2024), then 33.04% (2025)	Cancellation rate by month; cancellation timing
5	**Plan for the recurring September slowdown**	September revenue dip in 2023, 2024 and 2025	September revenue, occupancy and ADR vs other months

Areas to investigate based on descriptive evidence; no financial benefit is claimed.

Partial Years and Descriptive Metrics Limit How Far the Findings Reach

1. 2022 and 2026 are partial years; comparisons use full years 2023–2025.
2. Revenue and completed bookings need compatible date and record definitions.
3. Trends and cancellation rates do not prove causation.
4. Revenue per completed booking is descriptive, not ADR.

REPORTING PERIOD: APR 22, 2022 – JUL 31, 2026



NEXT ANALYSES

- Realized rates
- Booking mix
- Length of stay
- Cancellation timing

WHAT THIS PROJECT DEMONSTRATES

- ✓ End-to-end data preparation and modeling
- ✓ SQL-based data validation
- ✓ Power BI and DAX analytics
- ✓ Business-focused data storytelling
- ✓ Findings translated into management recommendations

From Raw Hotel Data to Actionable Business Insights.